

Customer Churn Analysis & Prediction Dashboard - Project Report

Executive Summary

This project focuses on analyzing customer churn patterns and generating actionable business insights using Power BI. The dashboard helps organizations identify churn drivers, monitor retention KPIs, and support proactive customer retention strategies.

Business Problem

Customer churn directly impacts revenue and growth. The objective is to understand why customers leave, identify high-risk segments, and provide predictive insights that enable data-driven retention initiatives.

Project Objectives

- Analyze customer demographics and behavior.
- Measure churn and retention rates.
- Identify factors influencing churn.
- Monitor customer lifetime value and revenue risk.
- Support strategic decision-making through interactive dashboards.

Data Preparation & Modeling

The project includes data collection, cleaning, transformation, handling missing values, feature engineering, relationship modeling, and optimization for analytical reporting.

Key Performance Indicators (KPIs)

- Total Customers
- Active Customers
- Churned Customers
- Churn Rate (%)
- Retention Rate (%)
- Revenue at Risk
- Customer Lifetime Value (CLV)
- Average Customer Tenure

Dashboard Components

1. Executive Overview Dashboard 2. Customer Segmentation Dashboard 3. Churn Trend Analysis
4. Revenue Impact Dashboard 5. Predictive Insights Dashboard 6. Retention Strategy Dashboard

DAX Measures & Analytics

Advanced DAX calculations are used to create dynamic KPIs, churn calculations, retention metrics, customer segmentation logic, and trend analysis.

Business Insights

- Identify customer groups with the highest churn probability.
- Understand revenue exposure from customer attrition.
- Detect behavioral patterns leading to churn.
- Improve retention through targeted interventions.

Technology Stack

- Microsoft Power BI
- Power Query
- DAX
- Data Modeling
- Business Intelligence & Analytics

Future Enhancements

- Machine Learning integration for churn prediction.
- Automated alert systems.
- Real-time dashboard refresh.
- Customer recommendation engine.

Conclusion

The Customer Churn Analysis & Prediction Dashboard transforms raw customer data into meaningful insights, enabling organizations to reduce churn, increase retention, and maximize long-term profitability.